Ocean-ICT

Investigation of the Effects of Ocean Acidification on **Ship Corrosion**

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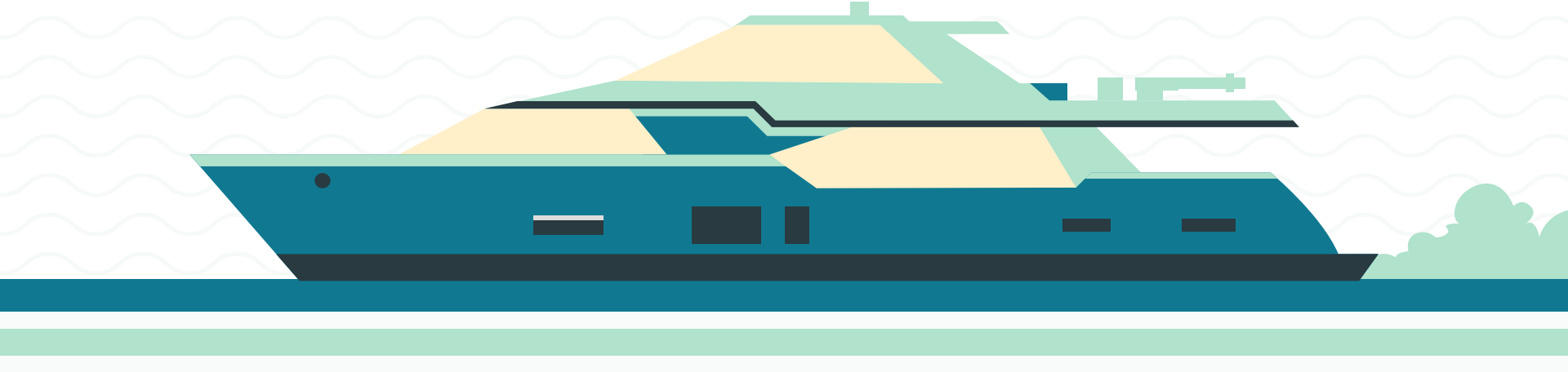
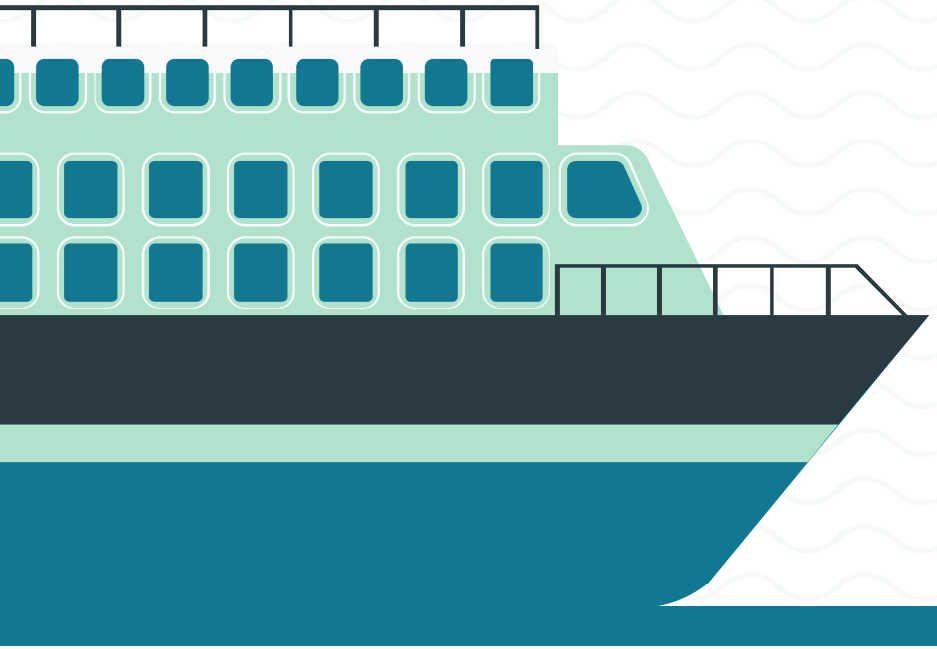
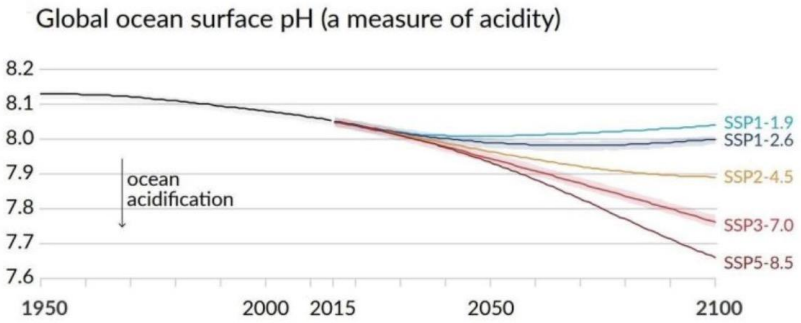
01

Introduction



Ocean Acidity has increased by **25%** since the 1700s





(Summary for Policymakers, 2023)
(US EPA, 2025)

This affects Marine Life

But what about Ships?

Our Aims

1

Prevent

Ship corrosion by developing an arduino-based device

2

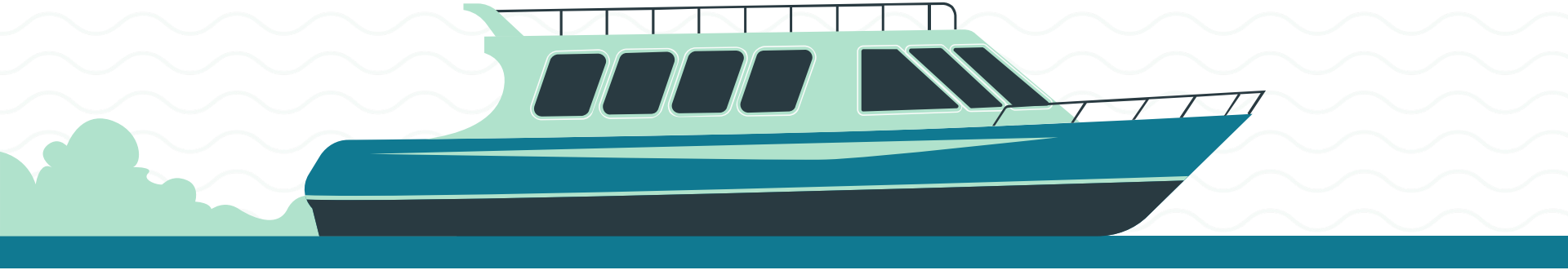
Understand

Environmental Impact of Ocean Acidification

3

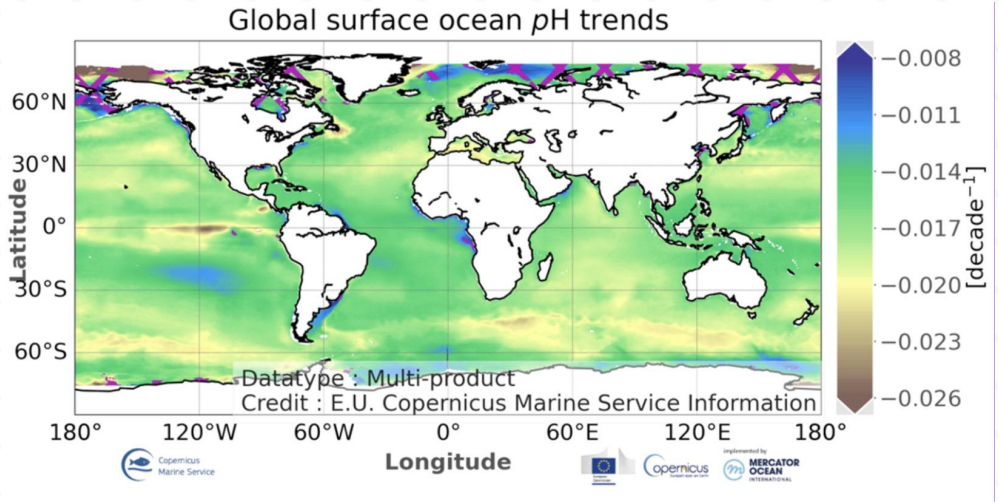
Inform

Future ship design



Localisation

Use regional pH data to customise the corrosion-prevention device for the location



02

Design

Considerations

Material

Aluminium alloy for main frame

Wiring

Must be water resistant

Sealant

Silicone, Waterproof tape, epoxy

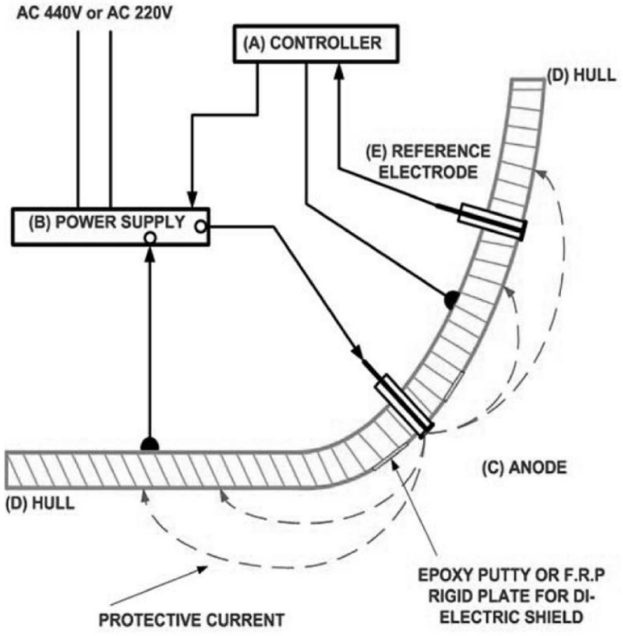
Buoyancy

Total weight < Buoyancy Force

Ocean Science

Reviewing Prev Research

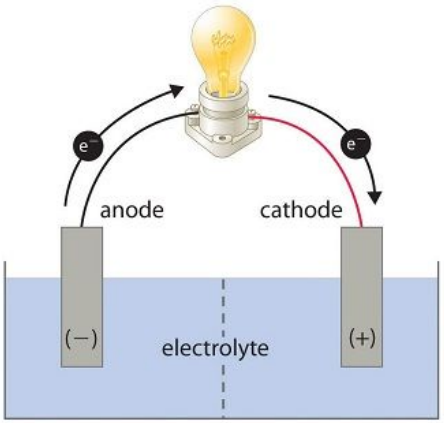
- ICCP (Impressed Current Cathodic Protection)
  - External power supply method for corrosion prevention
- Reference electrode
  - Detects corrosion potential of the ship's hull
- Controller
  - Sends the DC voltage signal to the power supply
- Power supply
  - Sends electrons from metal anode to hull (reduction electrode) via seawater (electrolyte).
- Saturated Calomel Electrode (SCE)
  - Cathodic protection is achieved when the hull voltage is below −0.77 V (vs. SCE).



ICCP was developed to prevent corrosion of ship in seawater by controlling their electrochemical potential.

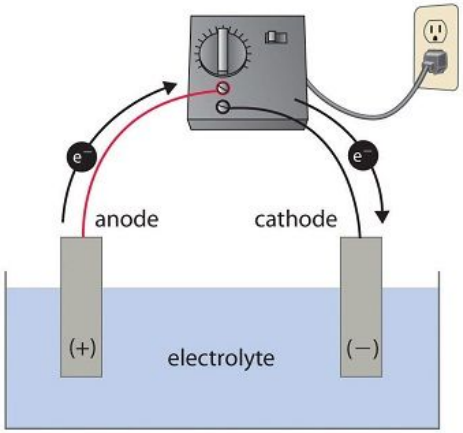
Chemistry

Galvanic Cell vs. Electrolytic Cell



GALVANIC CELL

Energy released by spontaneous redox reaction is converted to electrical energy.



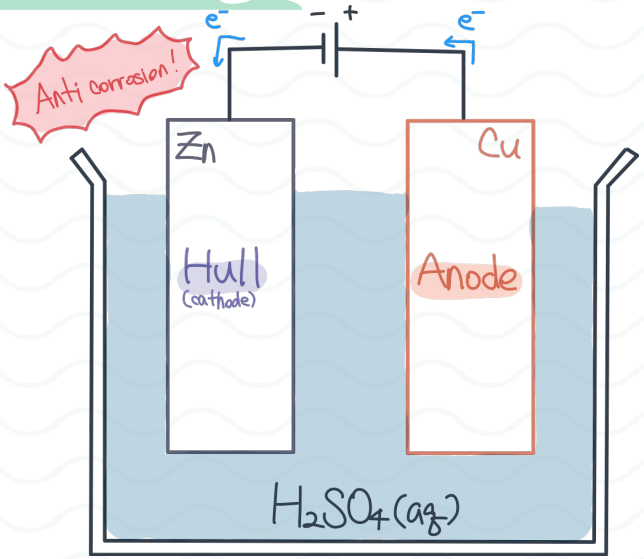
ELECTROLYTIC CELL

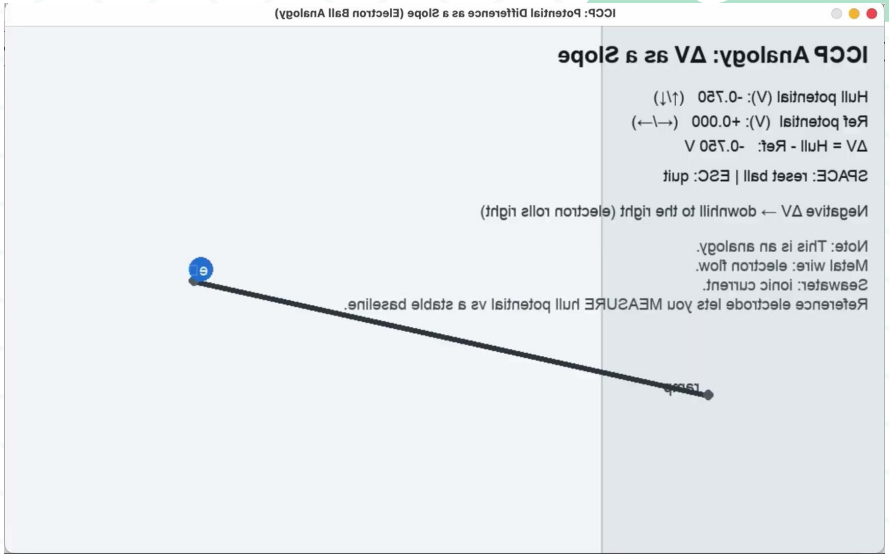
Electrical energy is used to drive nonspontaneous redox reaction.

In a **galvanic cell**, electrons flow **spontaneously** to produce electricity, whereas in an **electrolytic cell** a battery **pushes** electrons in the opposite direction to **drive reactions such as anti-corrosion.**

Chemistry

Setting up an Electrolytic Cell

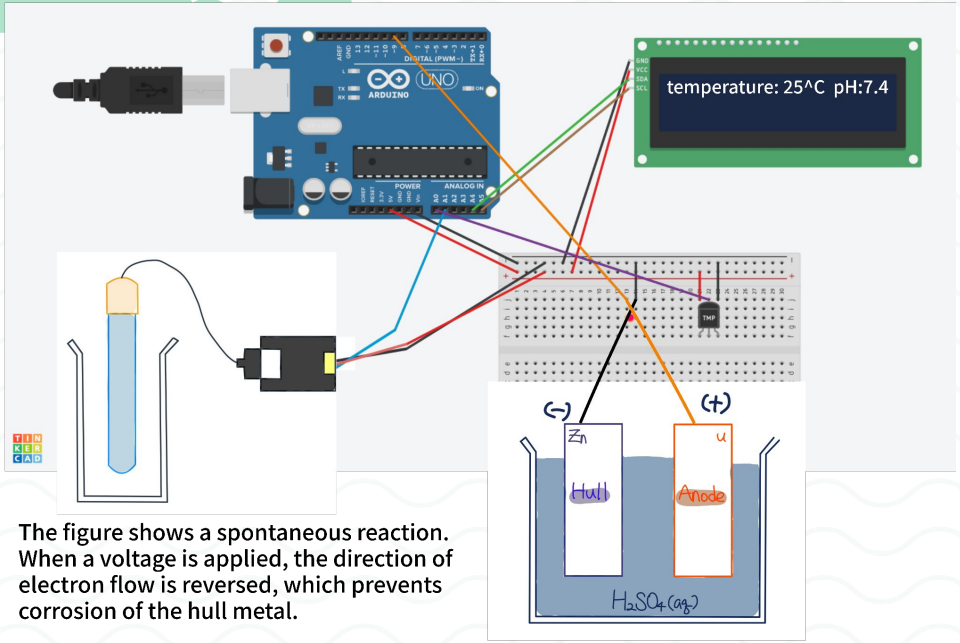




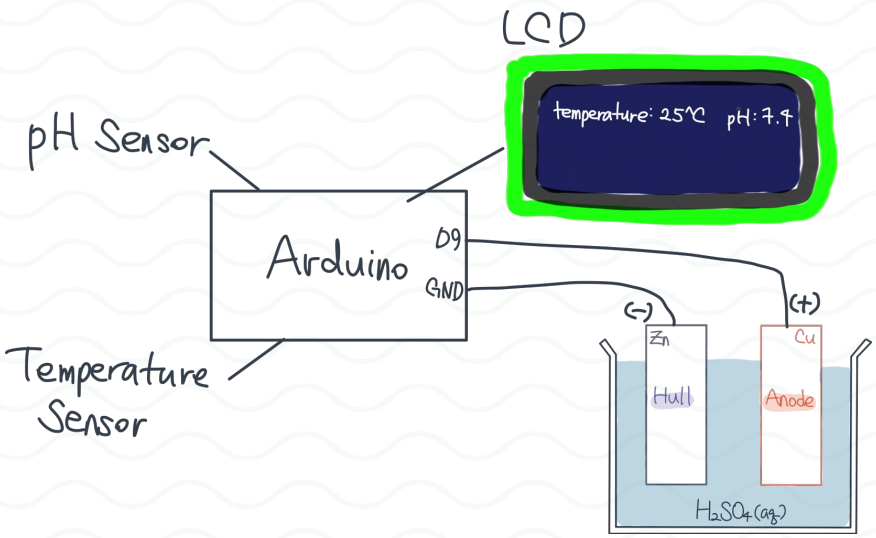
By applying external current to our **zinc–copper cell**, we force electrons to flow in the opposite direction, which **shifts the cell potential** (as shown in the simulation) and **prevents corrosion** of the hull.

Electronics

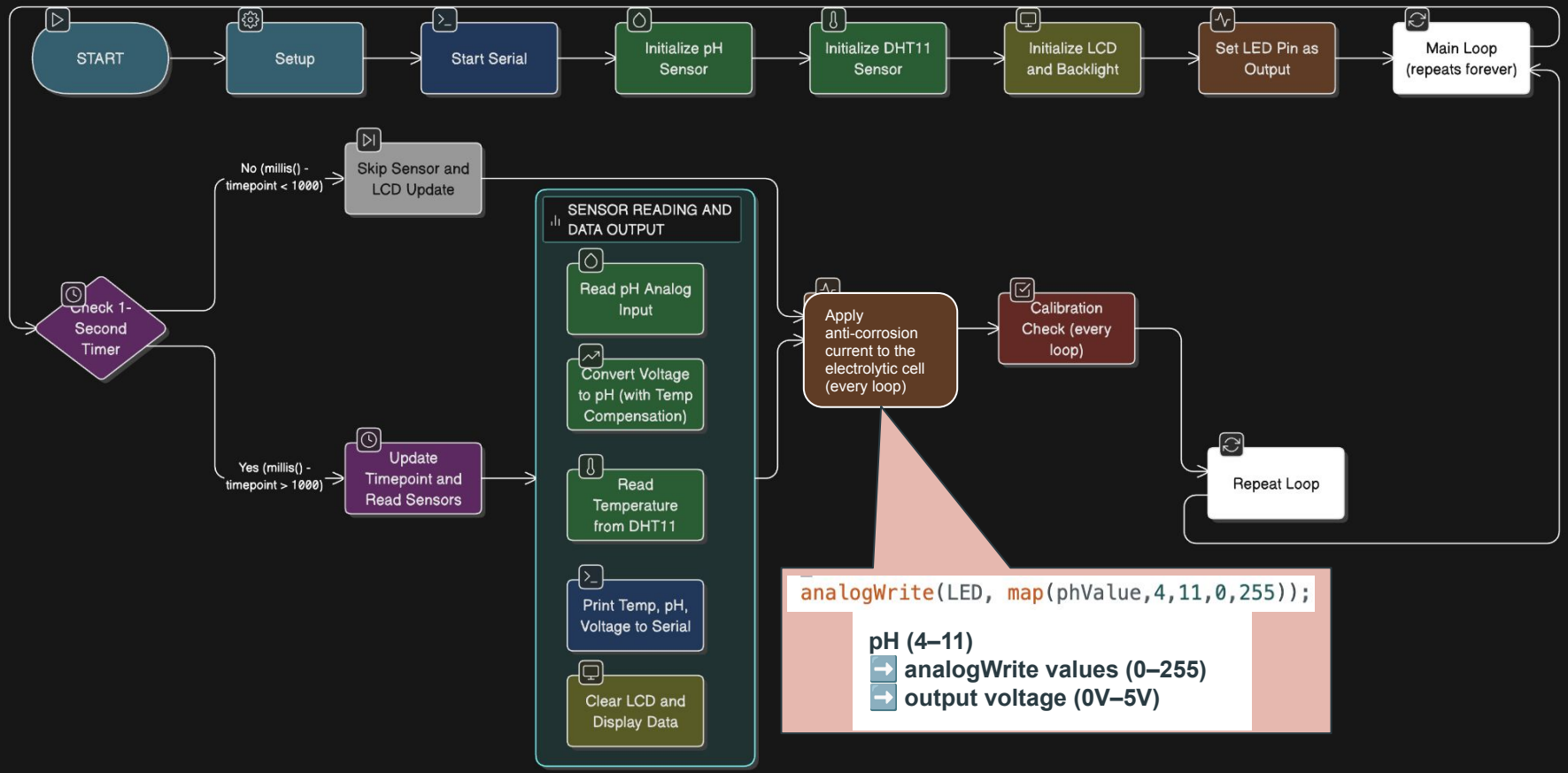
Designing Arduino Circuit



The figure shows a spontaneous reaction. When a voltage is applied, the direction of electron flow is reversed, which prevents corrosion of the hull metal.



By applying external current to our **zinc–copper cell**, we force electrons to flow in the opposite direction, which **shifts the cell potential** (as shown in the simulation) and **prevents corrosion** of the hull(zinc).

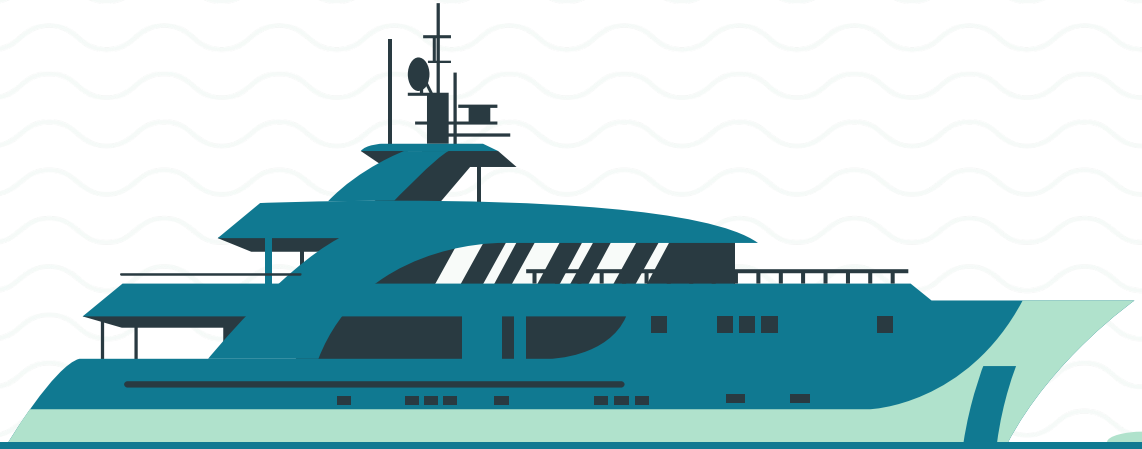


Flow Chart

eraser

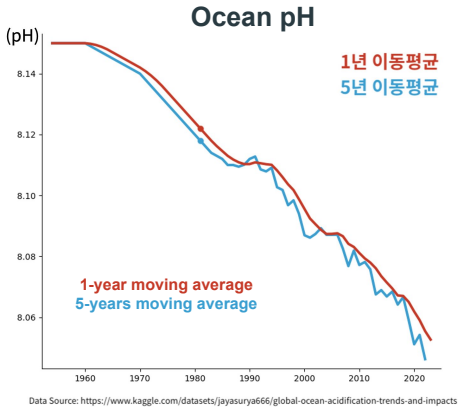
03

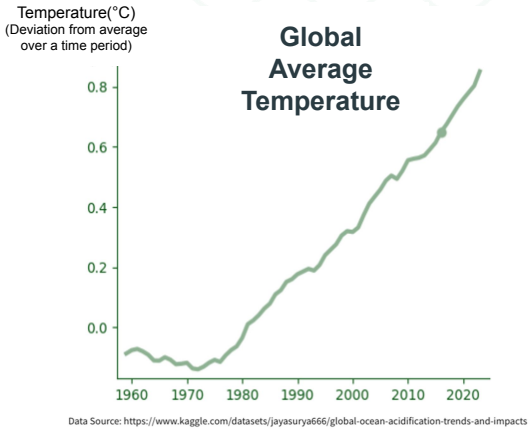
Implementation Result

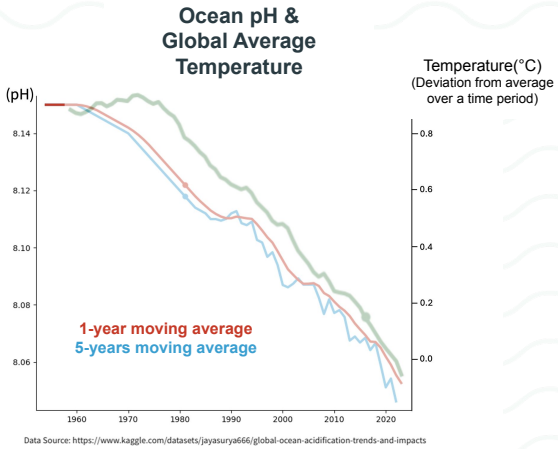


Information science

Data Analysis (Orange3)







Since the Industrial Revolution, global average temperature has increased, while ocean pH has decreased

Experiment Result

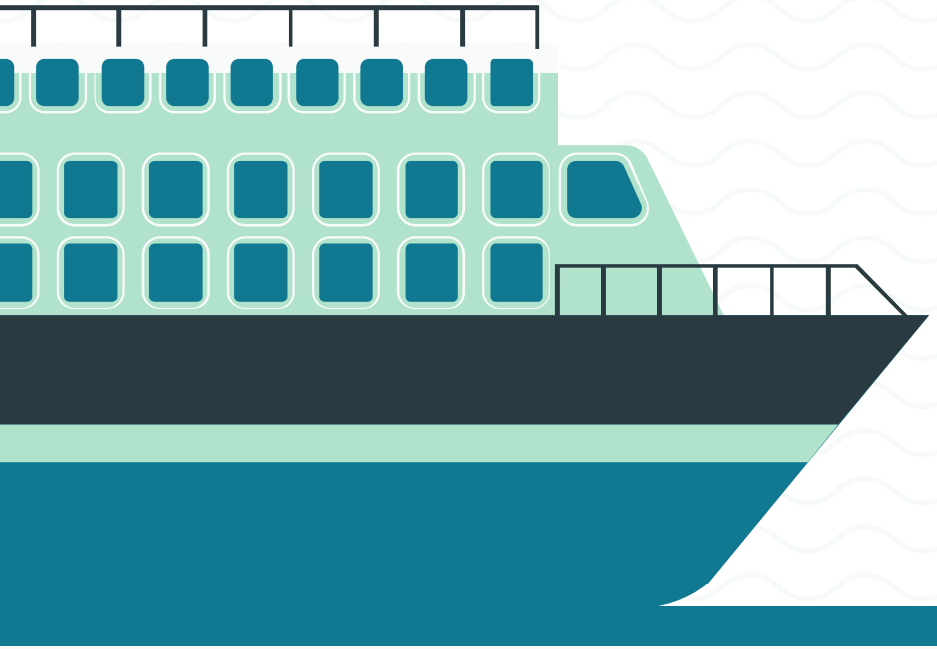
- Created a **prototype** for a vessel's **ICCP** system
- **Expected:** Hydrogen gas produced at Zn(zinc) electrode
- **Actual:** Hydrogen gas not produced at both electrode

We expected hydrogen gas to form at the zinc electrode(cathode), but even after applying an external current, we observed little to no visible change.

Learning from Our Failures

- **Insufficient current** to achieve effective ICCP behavior
- Arduino output **not designed** for electrochemical control
- Electrode potentials **not fully considered**
- The **pH sensor may not be suitable** for real-time measurement
- Possible **sensor contamination (pollution)**
- Hydrogen gas generation was **weaker** than expected

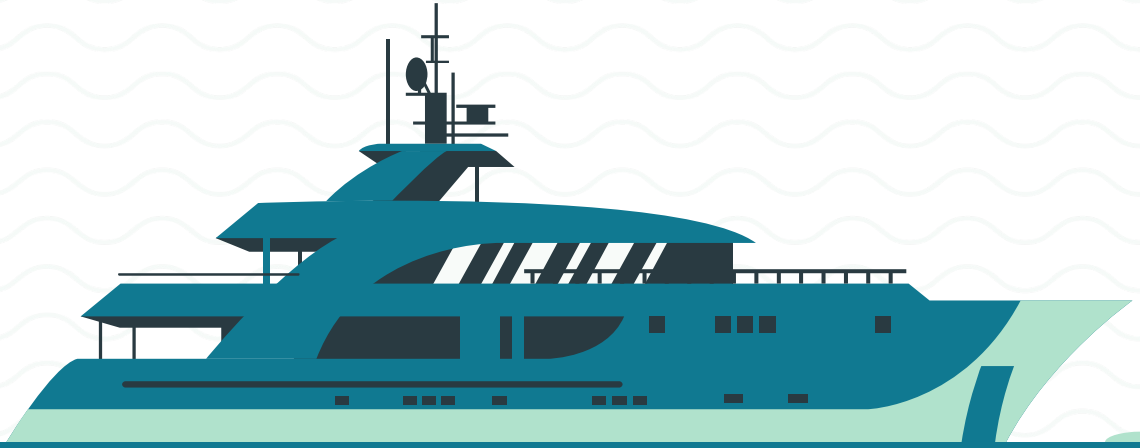
Further Research



- Analyze **Argo float data** for the ocean near Singapore
- Use a **higher-precision pH sensor**
- Explore electrical conductivity as an alternative method for estimating pH
- Investigate other cathodic protection methods (e.g., corrosion-resistant coatings)

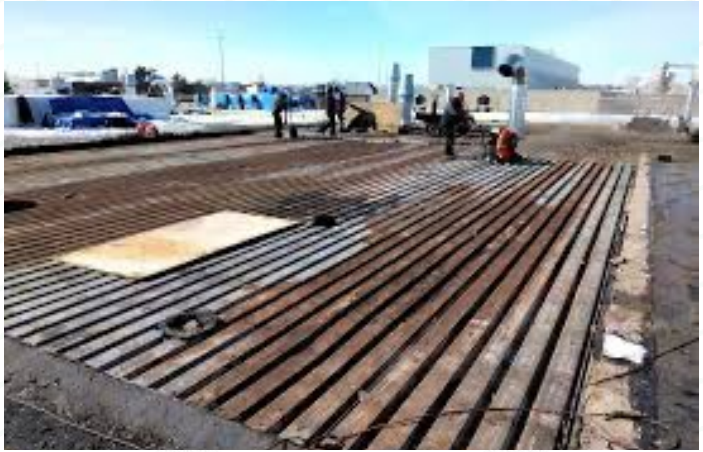
04

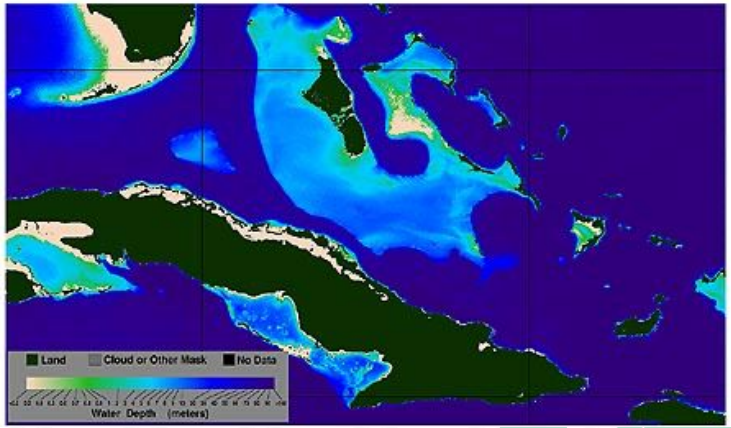
Application



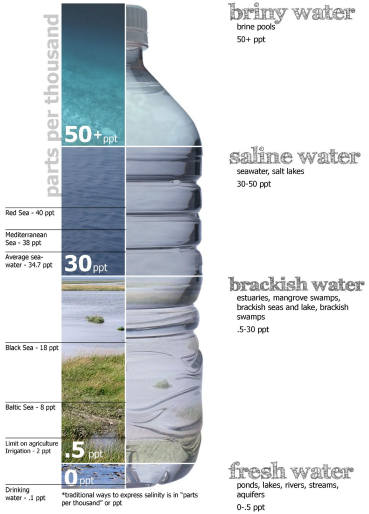
Corrosion of ships

- Use of specialised sensors and real time monitoring systems to measure corrosion rates
- experience has shown that even in the same ship corrosion rates vary significantly from location to location
- Corrosion of deck plating is affected by green water, rain and service water accumulation
- Water properties such as salinity, temperature, oxygen content, pH level and chemical composition can vary according to location and water depth





Real Life situations



Our prototype allows us to vary the following factors:

- pH -> mimic ocean acidification
- salinity -> electrical conductivity of seawater
- DO (dissolved oxygen) -> shallow vs deep water
- temperature -> tropical vs cold water

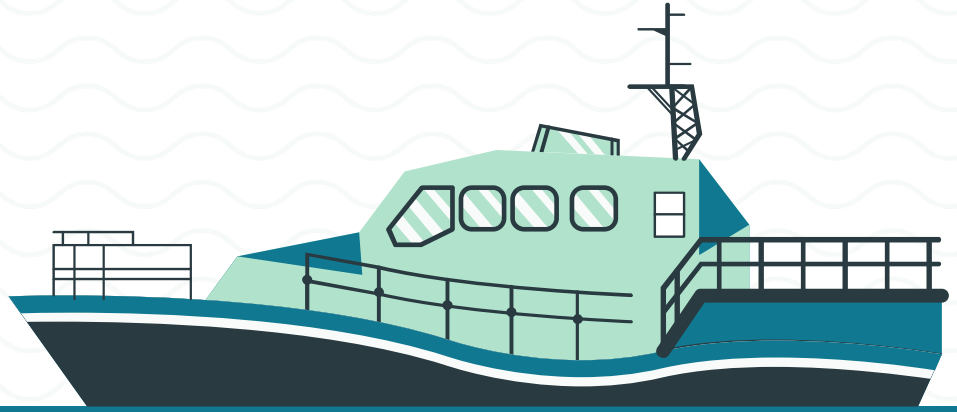
Afterwards, measure:

- corrosion severity
- ICCP DC current required, to reverse this natural process

https://www.sciencedirect.com/search?qs=estuaries

05

Relevance



How can this be applied to Singapore?

One of Singapore's main economic contributors is **Maritime Trade.**

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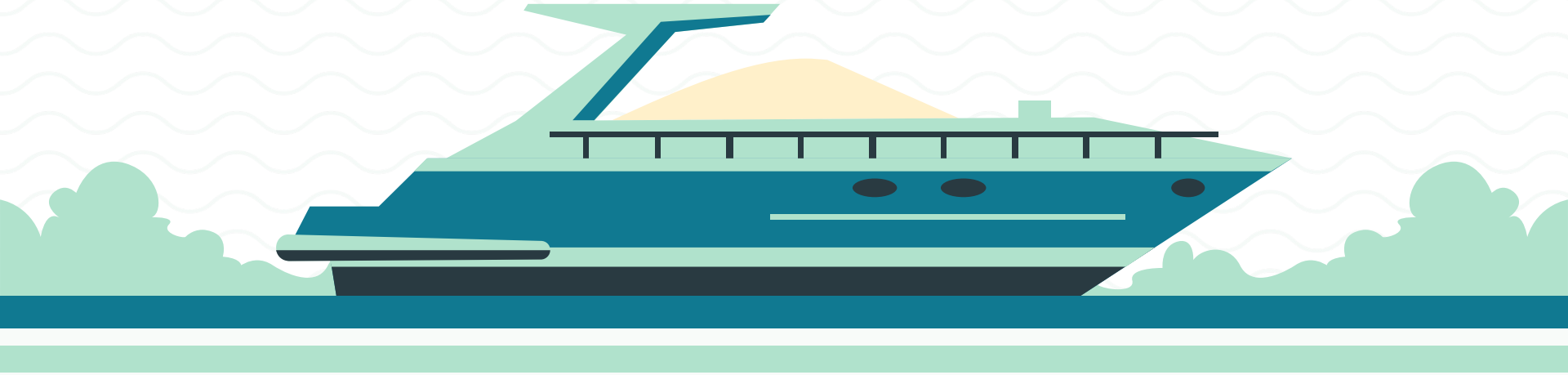
Vessels arrived at the Port of Singapore in 2024
(SINGSTAT, 2025)



Relevance

1

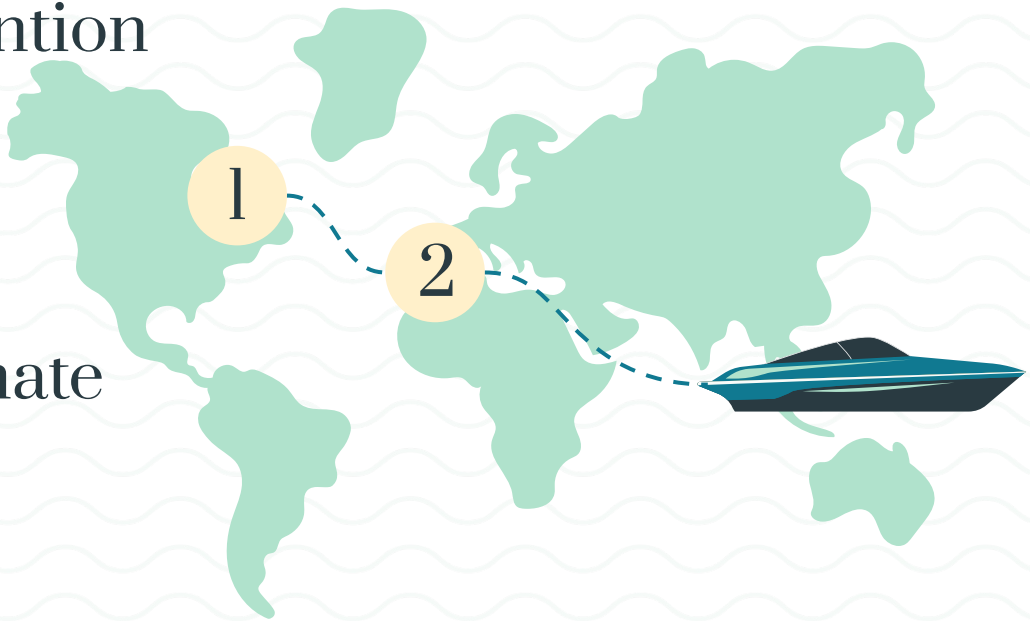
Corrosion Prevention

Current Methods include
expensive coating and blasting
(See Hup Seng, 2009)

2

Hot, Humid Climate

Singapore's Tropical weather
accelerates corrosion
(CorroSafe, 2024)



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